

MAT 425
FA 2025
Distributions

Name: _____

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What Is a Distribution?

- A distribution is a _____.
- Distributions _____ as _____ on test functions.
- A functional maps a set of functions to the _____.
- Test functions are _____ and have _____.

The Dirac delta function:

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = \underline{\hspace{2cm}}$$

What is this integral (for $d \neq 0$)?

$$\int_{a+|d|}^{\infty} f(x)\delta(x-a)dx = \underline{\hspace{2cm}}$$

Some Examples of Distributions

Problem 1. Dirac delta derivative: We will now find the weak derivative of the delta function. First, notice that:

$$\delta[f'] = \int_a^b \delta(x)f'(x)dx.$$

Integrate the RHS by parts to find that the weak derivative equals $-\delta[f']$.

Problem 2. Heaviside Problem: The Heaviside function is defined as:

$$H(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}$$

Show that the delta function is a weak derivative of H . This means that:

$$\int_a^b H(x)f'(x)dx = H(x)f(x)\Big|_a^b - \int_a^b \delta(x)f(x)dx$$

...by considering the integral above for the following three cases:

1. $a < b < 0$
2. $a < 0 < b$
3. $0 < a < b$